AKASH P

AspiringElectronicsand Communication Engineer

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EDUCATION

B.E - ELECTRONICS AND COMMUNICATION ENGINEERING (2021-2025)

PSNA COLLEGE OF ENGINEERING &TECHNOLOGY

CGPA- 7.95

HSC-2021

MSP SOLAI NADAR MEMORIAL HICHER SECONDARY SCHOOL

PERCENTAGE- 88%

SSLC-2019

MSP SOLAI NADAR MEMORIAL HICHER SECONDARY SCHOOL

PERCENTAGE- 86%

PROFILE

Passionate and detail-oriented Embedded Systems Engineer with hands-on experience in C, C++, Data Structures, and communication protocols such as UART, SPI, and I2C. Skilled in developing reliable and optimized hardware solutions using the LPC2129 microcontroller and Embedded C, with solid practical knowledge of FreeRTOS for real-time application development. Experienced in designing and implementing real-time embedded systems that enhance performance, reliability, and scalability. Strong understanding of circuit debugging, sensor integration, and system optimization. Eager to contribute technical expertise and creativity to innovative projects in embedded systems, aiming to deliver impactful, cost-effective solutions in the technology-driven industry.

PROJECTS

1. IOT - BASED UNDERGROUND CABLE FAULT DETECTION SYSTEM USING PIC16F877A MICROCONTROLLER - MAIN PROJECT

- **Concept Used:** Embedded C, IoT, Sensors, Fault Detection
- **Tool Used:** Arduino IDE, PIC C Compiler, ThingSpeak
- **Project Overview:**

This project presents an IoT-based underground cable fault detection system using the PIC16F877A microcontroller to locate faults accurately and efficiently. Traditional methods involve time-consuming manual inspections and excavation, leading to high costs and delays. The proposed system uses a potential divider circuit to detect voltage changes at the fault point, calculates the fault distance, and displays it on an LCD. It also transmits real-time fault data to the ThingSpeak cloud via the ESP8266 Wi-Fi module, enabling remote monitoring and data logging. The system is cost-effective, scalable, and capable of supporting smart grid infrastructure, thereby improving maintenance efficiency, reducing downtime, and enhancing the reliability and safety of underground power networks. Additionally, it allows utility providers to take quick action based on fault data, simplifies troubleshooting, improves operational transparency, and contributes to the digital transformation of electrical distribution systems in modern cities.

2.MOVIE TICKET BOOKING SYSTEM - MINI PROJECT

- **Concept Used:** C, Data Structure.
- **Tool Used:** Code::Blocks.
- **Project Overview:**

This C-based movie ticket booking system allows users to browse a wide selection of films, choose convenient showtimes, and reserve seats with real-time seat availability to prevent double bookings. After completing a booking, customers receive a detailed summary with their personal information and total cost. The system also includes administrative functions for modifying movie titles, adjusting ticket prices, updating showtimes, and managing seat cancellations, providing a seamless and enjoyable experience for both moviegoers and administrators.

SKILLS

- C
- C++
- Data Structures and Algorithms
- Embedded C-LPC2129 MC
- Protocols
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  - UART
  - SPI
  - I2C

SOFT SKILLS

- Leadership
- Communication
- Work Ethic
- Emotional Intelligence
- Adaptability Team
- Work Time
- Management

LANGUAGES

- English (Fluent)
- Tamil (Fluent)

3.IRRIGATION SYSTEM - MINI PROJECT

- **Concept Used:** Embedded C,LPC2129 Microcontroller.
- **Tool Used:** Keil µVision,Flash Magic.
- **Project Overview**

This advanced irrigation system, powered by the LPC2129 microcontroller, efficiently monitors soil moisture, temperature, and humidity through precision sensors. Based on real-time data, it automatically controls a water pump via a relay, optimizing water usage by turning the pump on or off as needed. The system features a user-friendly LCD display, showcasing sensor readings for easy monitoring of environmental conditions. This setup reduces water wastage and ensures crops receive ideal moisture, promoting healthy growth and maximizing yield. Its innovative design revolutionizes plant care, enhancing agricultural efficiency and sustainability.

INTERNSHIPS

PHOENIX SOFTECH (EMBEDDED C) **JUNE 2024 -JULY 2024**

During my internship at Phoenix Softech, I gained hands-on experience in Embedded C programming, working on real-time projects with the LPC2129 microcontroller. I extensively used communication protocols like UART, SPI, and I2C, deepening my understanding of embedded systems and their real-world applications. Collaborating on the design and development of embedded solutions improved my technical proficiency in coding and system integration. This experience broadened my knowledge and skills, equipping me to confidently address challenges in embedded technology and system development.

WORKSHOPS

- Attended a 2-day workshop on Hands-On Training in Real-Time IoT Projects.
- Attended a 2-day workshop on Hands-On Training in Voice Controlled Robot.

CERTIFICATIONS

- Certification in POP in Embedded System from **Cranes Varsity - Bengaluru**
- IOT Based Real Time Projects (Crystal Clear Technology). **(19-02-2024 To 20-02-2024)**
- Certificate for completing C Programming language. **(June 2023 -August 2023)**
- Certificate for completing C++ Programming language. **(June 2024 - July 2024)**
- Embedded C Internship Certificate (Phoenix Softech). **(June 2023 -August 2023)**